Towards a Seismic Vulnerability Database for Benefit-Cost Analysis Considering Retrofitting Interventions - an Example from Istanbul

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ABSTRACT

Earthquakes pose a significant risk to communities, largely due to vulnerable structures constructed before modern seismic design provisions. While retrofitting is a proven strategy to reduce seismic vulnerability and potential losses, most existing studies focus on individual buildings, offering limited insights into large-scale seismic risk mitigation. This study performs a scenario-based risk analysis for the city of Istanbul, which is amongst the most seismically active metropolitan areas worldwide. The work incorporates identifying the most vulnerable building classes, focusing on non-ductile reinforced concrete structures. Next, new vulnerability models are developed to capture different retrofitting scenarios featuring selected increments of ductility. A benefit-cost analysis is then carried out to demonstrate the effectiveness and economic viability of the different retrofitting scenarios for the city of Istanbul.